Project Report

Wildlife Information System (WIS)

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Prepared by:
Muhammad Ehtisham
Muhammad Abbas
Fahad Khan

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1. INTRODUCTION

1.1 Background

Wildlife means animals that are living naturally in forests, mountains, deserts, local forms and in other environments like elephants, lions, wolves, dogs cows etc. Those animals are very important because they helps to maintain the balance of nature. like predators like as wolves controls the population of other animals, and the birds helps plant to grow by spreading seeds and pollinating flowers. much information about wildlife is collected around the world. This information includes Animal, population number, Movements and migrations routes, Animal's behavior. DNA and genetic information. Health and disease records. However, and There will be real time simulations real time movements its health notification etc. the data is stored in many different places. Different organizations, universities, researchers, and other government departments can easily collect data separately. and There is no single system where all wildlife information can be found. In Pakistan and many other countries, wildlife data is usually stored in reports, spreadsheets, notebooks, and paper records. Because of this, it become difficult for researchers, Doctors, and government and other agencies to access and use the information.

Our System the Wildlife Information System (WIS) is proposed for solving this problem. WIS is a centralized system that can stores all wildlife data in one organized database. Users can easily access and share information through an API, making wildlife research and conservation more efficient. And shows the real time simulation of animals and health records. So Owner can quickly take step.

1.2 Existing System

Currently, the data of wildlife in most countries is managed in these the following ways:

- Government forest departments maintain their own paper base or basic spreadsheets or excel file records of wildlife populations and incidents.
- Researching universities collect the data during research or field trips and store or save it in local databases that are not accessible to the public.
- Other organizations like International as "WWF" and "IUCN" maintain global databases, but they are limited and they have not included local or regional level details.
- And Some IoT based systems which tracks animal movement using GPS collars, but that data kept private and which did not shared in public platforms.

The major weakness of all these approaches is that they do not talk to each other. There is no standard system that collects, organizes, and shares wildlife data openly.

1.3 Issues & Motivation

The following problems motivate the development of the Wildlife Information System:

- Data Isolation: Wildlife data exists in silos. Researchers, forest departments, and NGOs each have their own records but cannot share them.
- No Standard Format: Data is stored in different formats like Excel files, PDF reports, handwritten notes etc. we are making it impossible to combine or compare across organizations.
- Lack of Open Access or Source: Most wildlife databases are closed. Researchers must physically contact organizations and wait weeks to get data they need for their studies.
- Missed Opportunities for Analysis: Because data is fragmented, important patterns such as disease outbreaks, migration route changes, or population decline are often noticed too late.
- Waste of Research Resources: Multiple researchers sometimes collect the same data independently because they do not know it already exists elsewhere. Which waste their time.

1.4 Problem Statement

There is no single system in Pakistan where all the wildlife information is stored or shared. The different universities, research institutions and government institutions have but they keep their own records separately or private, because of this. It is most difficult to find information about animals like their populations, their behavior, DNA and Their Health. This creates many problems Researchers may collect same data repeatedly because they don't know it already exists. And also, the most important information can be missed, and govt agencies may take decisions without complete or accurate data.
The Wildlife Information system solves these problems. It will store wildlife information in one central database. And Authorized users can access this information through secure API and they can use it for research, wildlife monitoring, conservation projects and they can use for decision making.

1.5 Scope

The Wildlife Information System will cover the following:

- It will keep all the wildlife information in one central database. The information will include like animal profile, Population, movement tracking, DNA records, health, their behavior and environment data.
- There will be a backend system that will validate, organize, and manages all incoming data from different sources, like field researchers, IoT tracking devices, and lab results.
- It will include an API. A RESTful API layer that allows external users and applications to send queries and receive structured wildlife data in standard format like JSON.
- The User roles include Data Entry Operators, Researchers, Government Officials, and System Administrators, each will have different level of access.
- There will be a basic web interface in which can search and can enter data manual.

The following are explicitly outside the scope of WIS:

- This system will not install GPS collar or tracking device, manufacturing, or design. It will only store data transmitted from existing devices.
- Real-time live animal tracking maps or animated movement visualization on a geographic interface (this may be a future Phase 2 extension).
- This system will not provide any real time data or tracking or animated maps showing animal movements. But this may be a future plan in version 2.0 extension
- This system will not provide any mobile application for android or iOS. But the other developers may build these separately using the WIS public API.
- This system will not use AI or ML for identifying animal from images or sensor data. This system stores records but does not perform intelligent pattern recognition.
- This system will not connect directly with international database like IUCN Red list or Cities in v1.0 but may connect in later versions.
- This system will not include financial, HR, payroll, or administrative management features. It's mainly focuses on ecological and animal data.
- This system will not include offline data collection tools or special devices for field use in remote areas where the internet is not available.
- This system will not provide automatic alerts for diseases outbreaks, environmental risks or illegal hunting etc.
- This system will not support multiple languages. this system will operate in English only in v1.0.
- The system will not handle legal reporting wildlife permits or enforcement tracking systems.

1.6 Project Management

This project will be managed with a structured approach and everything is clearly defined in phases, responsibilities, and timelines.

Phase	Activity	Duration
Phase 1	Requirements gathering and analysis	2 weeks
Phase 2	System design (DB schema, API design)	2 weeks
Phase 3	Backend development and database setup	3 weeks
Phase 4	API development and testing	2 weeks
Phase 5	Documentation and final review/Testing	1 week

We will use the tools like GitHub for version control, WhatsApp and Physical or online meeting for team communication, and Google Docs for shared documentation. Weekly meetings every Saturday to check progress. And discuss.

1.7 Technological Overview

Component	Technology
Backend	Nodejs with Expressjs
Database	PostgreSQL
API Type	RESTful API (JSON responses)
Authentication	JWT (JSON Web Tokens)
Frontend (basic)	HTML, CSS(Tailwindcss), JavaScript, Reactjs
Version Control	Git / GitHub
Hosting (future)	Cloud (AWS or DigitalOcean)
Data Format	JSON and CSV for exports

1.8 Goals & Responsibilities

Primary Goal: To build a centralized wildlife information database that is accurate, and do simulation to view real time movements and health status, well organized, and accessible through an API for research and analysis purposes.

Secondary Goal: Encourage data sharing among wildlife organizations. and we will also provide a platform that supports conservation and research decision making easily. Also we will implement and build a scalable foundation that can grow with more data and users over time. Which can handle more user at a time.

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1.9 Process Model

we will use DevOps practices for development, testing, integration and deployment. DevOps combines two thing development and operations for improve collaboration, automation, and repetitive tasks. And for maintain the software quality and ensure system reliability, Version control, automated testing, continuous integration and deployment practices will be used. To manage code changes maintain stability. Also, we will use Model View Controller (MVC) architectural pattern for organize. Models manage app data and business logic. The View will provide UI where user Interact. And the controller will handle user request and coordinate communication between the model and view. Separate parts will make the system easier to develop, testing, and maintain and scaling/extend.

1.10 Project Management System

The team will use GitHub Projects to manage the work of the Project. All Tasks will be organized in a board format with three columns:

1. **To Do:** The tasks that need to be started
2. **In Progress:** The tasks that are currently being worked on.
3. **Done:** The tasks that have been completed

Each team member is assigned specific issues or tasks on the GitHub repository. Which will help us in tracking progress and make sure every task is completed on Time. And later then discuss what each member did.